Lecture Notes · Geometry · Session 1 — The affine space: points, vectors, lines and planes

All the session material with the full formalization: the axioms of the affine space and their consequences, the three forms of the equation with their equivalence proved, and the theorem identifying affine subspaces with solution sets. Exercises and a Going further section at the end.

1.1 The affine space

Vector spaces have a privileged point: the 0. The plane of classical geometry does not — no point is special. What there *is* between two points is the **displacement** connecting them. The structure that captures this:

Definition 5.1.1 (affine space). An **affine space** over a vector space V (over K) is a nonempty set $\mathbb{A}$ of **points** together with an operation

$$\mathbb{A} \times V \to \mathbb{A}, \qquad (P, v) \mapsto P + v,$$

satisfying: 1. $P + 0 = P$ for every point P; 2. $(P + v) + w = P + (v + w)$ (translating twice is translating by the sum); 3. for each pair of points (P, Q) there exists **a unique** vector, denoted $\vec{PQ}$, with $P + \vec{PQ} = Q$.

The **dimension** of $\mathbb{A}$ is that of V. **Model:** $\mathbb{A}^n$ = the points of K^n, with $V = K^n$ acting by addition of tuples; the axioms are immediate.

Remark 5.1.1b (adding points, no; subtracting them, yes). Note the division of roles the definition imposes: **adding two points makes no sense** — the only thing one adds to a point is a *vector* — but **subtracting them does**, and the result is a vector: it is exactly $\vec{PQ}$, the unique vector of axiom 3. Many texts write it $Q - P$; here we shall always use the arrow, $\vec{PQ}$.

Proposition 5.1.2 (consequences). In every affine space: 1. $\vec{PP} = 0$, and $P + v = P \Rightarrow v = 0$. 2. **(Chasles)** $\vec{PQ} + \vec{QR} = \vec{PR}$. 3. $\vec{QP} = -\vec{PQ}$. 4. **(parallelogram rule)** $\vec{PQ} = \vec{P'Q'} \iff \vec{PP'} = \vec{QQ'}$.

Proof. (1) $P + 0 = P$ by axiom 1, and the vector taking P to P is unique (axiom 3): $\vec{PP} = 0$; the second claim is the same uniqueness. (2) $P + (\vec{PQ} + \vec{QR}) = (P + \vec{PQ}) + \vec{QR} = Q + \vec{QR} = R$ (axiom 2), and by uniqueness that vector is $\vec{PR}$. (3) Chasles with $R = P$: $\vec{PQ} + \vec{QP} = \vec{PP} = 0$. (4) By Chasles twice: $\vec{PQ'} = \vec{PP'} + \vec{P'Q'} = \vec{PQ} + \vec{QQ'}$; equating $\vec{P'Q'} = \vec{PQ}$ amounts to equating $\vec{PP'} = \vec{QQ'}$ (cancelling in V). ■

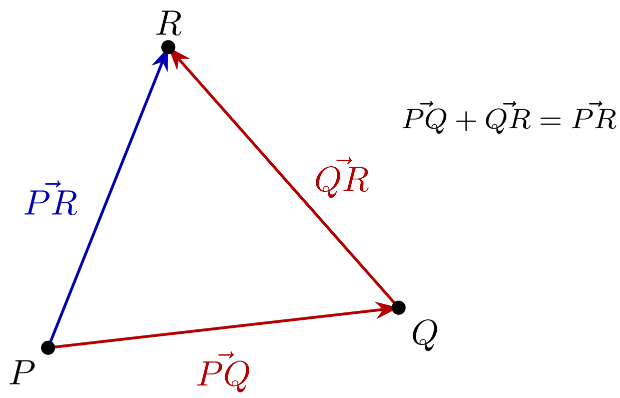

Figure 1: Chasles' relation: $\vec{PQ} + \vec{QR} = \vec{PR}$ (chained vectors), and $\vec{QP} = -\vec{PQ}$.

Affine = linear + reference point

Proposition 5.1.3. Once a point $O \in \mathbb{A}$ (an **origin**) is fixed, the assignment $P \mapsto \vec{OP}$ is a **bijection** between the points of $\mathbb{A}$ and the vectors of V.

Proof. Injective: if $\vec{OP} = \vec{OQ}$, then $P = O + \vec{OP} = O + \vec{OQ} = Q$. Surjective: $v = \overrightarrow{O(O+v)}$ by axiom 3. ■

> The identification **depends on the choice of** O: with another origin O', the vector associated with P changes ($\vec{O'P} = \vec{O'O} + \vec{OP}$, Chasles). That is why *point* and *vector* are different concepts: identifying them requires a choice. This is the precise content of the slogan **"affine = linear + reference point"**.

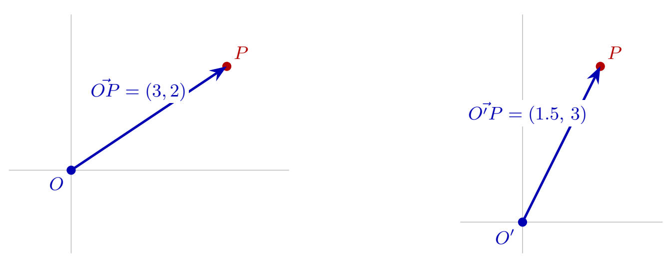

Figure 2: Drag the origin O (in the interactive version): the point P does not move, but its vector $\vec{OP}$ — and hence its coordinates — changes with the choice of O.

[see interactive version]

Definition 5.1.4 (frame of reference and coordinates). A **frame of reference** of $\mathbb{A}$ (dimension n) is a pair (origin O, basis $\{b_1, \dots, b_n\}$ of V). The **coordinates** of the point P are the coordinates of the vector $\vec{OP}$ in that basis — unique, by Proposition 4.3.18 of Linear Algebra.

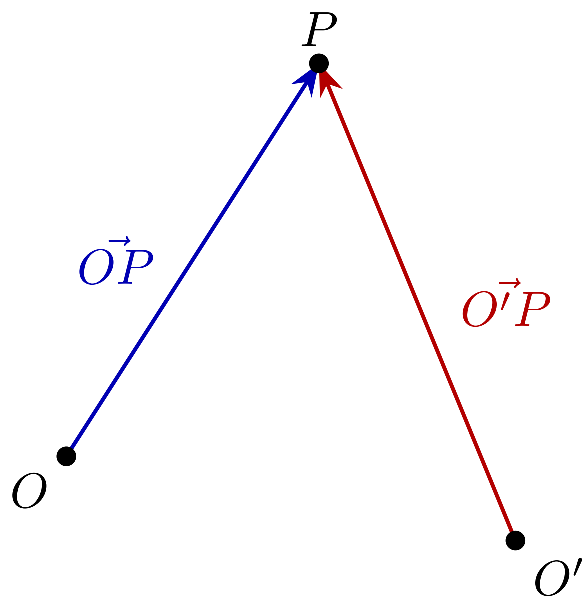

Figure 3: The base point is not intrinsic: the same P has different vectors $\vec{OP}$ and $\vec{O'P}$ depending on the origin, and hence different coordinates.

1.2 Affine subspaces; lines and planes

Definition 5.1.5. An **affine subspace** of $\mathbb{A}$ is a set of the form

$$P + W = \{\, P + w : w \in W \},$$

with P a point and $W \subseteq V$ a vector subspace, called the **direction space**. Its **dimension** is $\dim W$: **points** (dim 0), **lines** (dim 1), **planes** (dim 2), **hyperplanes** (dim $n-1$).

Proposition 5.1.6 (the direction space is intrinsic; the base point is not). If $P + W = P' + W'$, then $W = W'$ and $P' \in P + W$. Conversely, $P + W = P' + W$ for **any** $P' \in P + W$.

Proof. If $P' \in P + W$, then $\vec{PP'} \in W$ and $P' + W = P + (\vec{PP'} + W) = P + W$ (because $\vec{PP'} + W = W$: translating a subspace by one of its own vectors leaves it unchanged — closure). For the first part: $W = \{\vec{XY} : X, Y \in P + W\}$ (differences of points of the set, via Chasles), which depends only on the set. ■

Definition 5.1.7 (line and plane, vector form). The **line** through P with **direction vector** $v \neq 0$: $r = P + \operatorname{span}(v) = \{P + tv : t \in K\}$. The **plane** through P with **independent** direction vectors u, v: $\pi = P + \operatorname{span}(u, v) = \{P + su + tv : s, t \in K\}$.

It is **the same definition** with k independent directions, $P + \operatorname{span}(v_1, \dots, v_k)$: $k = 1$ gives the line, $k = 2$ the plane, $k = n - 1$ the **hyperplane** and $k = n$ the whole space. All that changes is how many parameters are needed.

Remark 5.1.7b (why those hypotheses). Both conditions are needed for the name to be the right one: with $v = 0$ the set $P + \operatorname{span}(v)$ collapses to the point P (dimension 0, not 1), and with u, v proportional $\operatorname{span}(u, v)$ has dimension 1, so that $P + \operatorname{span}(u, v)$ is a line and not a plane. They are not decoration: they are what guarantees the announced dimension.

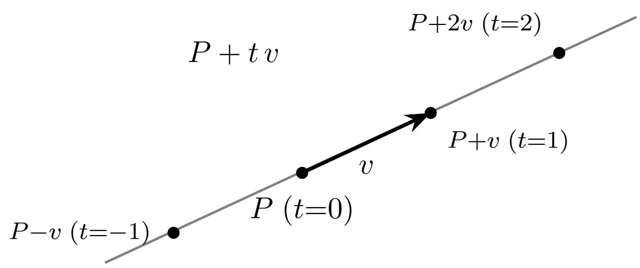

Figure 4: The line $P + tv$: as t varies, the whole line is traced out in the direction of the direction vector v (here $t = -1, 0, 1, 2$).

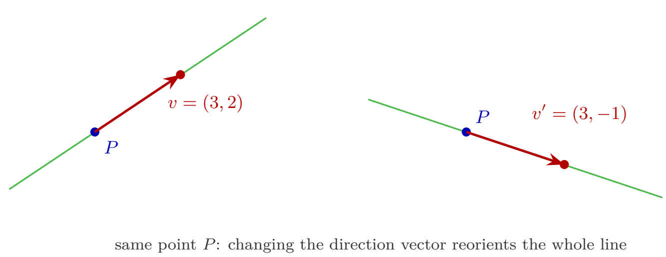

Figure 5: Interactive version: drag the tip of the direction vector to see how, as v changes, the whole line reorients.

[see interactive version]

The three forms and their equivalence

In coordinates (once a frame of reference is fixed in $\mathbb{A}^n$):

1. **Vector form:** $X = P + t_1 v_1 + \cdots + t_k v_k$, with $v_1, \dots, v_k$ **independent** (and then $k = \dim$ of the subspace, Definition 5.1.5: the v_i are a basis of the direction space).
2. **Parametric:** the above, coordinate by coordinate (n equations with k parameters).
3. **Implicit:** the set as the solutions of a linear system $Ax = b$.

Theorem 5.1.8 (affine subspaces = solution sets). In $\mathbb{A}^n$: 1. The solution set of **any** consistent system $Ax = b$ is an affine subspace, with direction space the kernel of A (and dimension $n - \text{rank}A$). 2. Conversely, **every** affine subspace $P + W$ is the solution set of some consistent linear system.

Proof. (1) It is the Theorem "general = particular + kernel" (Linear Algebra, Theorem 4.4.7) reread with Definition 5.1.5: solutions $= x_p + \ker A$; the dimension is Theorem 4.4.6 of Linear Algebra.

(2) It suffices to find a **homogeneous** system whose solutions are exactly W (of dimension k); afterwards the right-hand side is adjusted with $b := A \cdot P$ to translate to $P + W$ (the solutions of $Ax = AP$ are $P + \ker A$ by part 1). Let $\{w_1, \dots, w_k\}$ be a basis of W and M the $k \times n$ matrix with rows w_i. The kernel of M has dimension $n - k$ (the rank of M is k: independent rows); let $\{z_1, \dots, z_{n-k}\}$ be a basis of it and A the matrix with rows z_j. Then:

- each w_i is a solution of $Ax = 0$: the condition $Aw_i = 0$ says $\sum_l (z_j)_l (w_i)_l = 0$ for each j, which is exactly $Mz_j = 0$ read backwards (the expression is **symmetric** in w_i and z_j). Hence $W \subseteq \ker A$.
- $\dim(\ker A) = n - \text{rank}A = n - (n - k) = k = \dim W$. And a subspace contained in another of the same finite dimension coincides with it. *Indeed:* a basis of W is an independent family of k elements inside $\ker A$; by Lemma 4.3.18b(1) of Linear Algebra it extends to a basis of $\ker A$ — but that basis also has k elements ($\dim \ker A = k$), so the extension adds nothing: the basis of W was already a basis of $\ker A$, and $W = \text{span} = \ker A$. ∎

Method 5.1.9 (converting between forms).

- *Parametric → implicit:* **eliminate the parameters**. In general, with $X = P + t_1 v_1 + \cdots + t_k v_k$, one views it as a system whose unknown is the vector of **all** the parameters $t = (t_1, \dots, t_k)$:

  $$Vt = X - P, \qquad V = (\, v_1 \mid \cdots \mid v_k \,) \text{ (the directions, as columns)},$$

  and the implicit equations are the consistency condition (Rouché–Capelli), in which no parameter appears any more:

  $$\text{rg}\,[\, V \mid X - P] \;=\; \text{rg}\, V \;=\; k$$

(the last equality because the v_i are independent). *(In practice, with one or two parameters: solve and substitute, or reduce to echelon form.)* **Note** that this V is **not** the matrix A of the implicit equations: they are two different systems, one with the parameters as unknowns and one with the coordinates.

- *Implicit → parametric:* **solve the system** and parametrize — literally Method 4.4.4 of Linear Algebra.

> **Example 5.1.10.** The line of $\mathbb{R}^3$ through $P = (1, 0, 2)$ with direction vector $v = (1, 1, -1)$:
>
> - *Vector form:* $(x, y, z) = (1, 0, 2) + t(1, 1, -1)$.
> - *Parametric:* $x = 1 + t$, $y = t$, $z = 2 - t$.
> - *Implicit:* eliminate t. Seen **as a system in the unknown** t (with x, y, z playing the role of right-hand side), the parametric equations read $Vt = X - P$ with $V = (1,\ 1,\ -1)^{\mathsf{T}}$ and $X - P = (x-1,\ y,\ z-2)^{\mathsf{T}}$, and the consistency condition $\operatorname{rank}[V \mid X - P] = 1$ says that the three coordinates on the right must be proportional to $(1, 1, -1)$, that is, $x - 1 = y$ and $z - 2 = -y$: the **implicit** equations $x - y = 1,\ \ y + z = 2$. Written as $Ax = b$, here
>
> $$A = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 1 \\ 2 \end{pmatrix},$$
>
> with $\operatorname{rank} A = 2$, so the dimension of the solution set is $n - \operatorname{rank} A = 3 - 2 = 1$ ✓ — a line, which is the k we started from. **This check is worth doing every time**: it catches translation errors. *(In practice one does not write the matrix down: solve for $t = y$ in the second equation and substitute in the other two. It is the same computation, done in your head.)*

Hyperplanes. A single **nondegenerate** linear equation — that is, with some $a_i \neq 0$ — $a_1x_1 + \cdots + a_nx_n = b$ defines an affine subspace of dimension $n - 1$ (rank 1): a **hyperplane**. *(The hypothesis matters: if all the a_i vanish, the equation becomes $0 = b$ and describes no hyperplane at all — it is empty if $b \neq 0$, and the whole of $\mathbb{A}^n$ if $b = 0$.)* Lines in the plane and planes in space are the cases $n = 2, 3$; the reading "each equation of a system carves out a hyperplane, and solving is intersecting them" (Session 1 of Linear Algebra) is now fully formalized.

Exercises

1. ★ For the line of $\mathbb{R}^3$ through $P = (0, 1, 1)$ and $Q = (2, 1, 3)$: find a direction vector, write the three forms and check that Q satisfies them.
2. ★ Convert to parametric form (by solving) the plane $x - 2y + z = 4$ of $\mathbb{R}^3$, and to implicit form (by eliminating parameters) the plane $(x, y, z) = (1, 0, 0) + s(1, 1, 0) + t(0, 1, 1)$.
3. ★ The system $\{x + y - z = 1;\ 2x - y = 0\}$ is consistent. Describe its solution set as an affine subspace: base point, direction space (a basis) and

dimension.

4. ★ **(Point vs. vector / affine vs. linear.)** The line $y = x + 1$ of $\mathbb{R}^2$: is it a vector subspace? is it an affine subspace? Give its direction space and exhibit two different choices of base point (Proposition 5.1.6).
5. Prove with Chasles: the midpoint M of P and Q (defined by $\vec{PM} = \frac{1}{2}\vec{PQ}$, with $K = \mathbb{R}$) satisfies $\vec{MP} + \vec{MQ} = 0$.
6. Check the three axioms of Definition 5.1.1 in the model $\mathbb{A}^n$.
7. Let r be the line through P with direction vector v. Prove that $Q \in r \iff \vec{PQ}$ is proportional to v, and deduce a criterion for **three points to be collinear**.
8. In Theorem 5.1.8.2, follow the construction with $W = \operatorname{span}((1,1,0),(0,1,1)) \subseteq \mathbb{R}^3$ and obtain implicit equations for the plane $(1,2,3) + W$.

Going further and references

Going further (affine combinations and barycenters). Although "adding points" makes no sense, the combinations $\sum \lambda_i P_i$ with $\sum \lambda_i = 1$ do (the result does not depend on the origin used to compute them): these are the **affine combinations**, and with them come barycenters, segments and convexity. See de Burgos, *Álgebra lineal y geometría cartesiana*, or Audin, *Geometry*, ch. I.

References. Axiomatic affine space: de Burgos; Audin, ch. I. Lines, planes and the forms of the equation: de Burgos; Hernández, *Álgebra y geometría*. Affine subspaces as solutions: Hefferon, One.I (implicitly), and the Linear Algebra notes, Session 4.

Lecture Notes · Geometry · Session 2 — Relative positions

The general method (intersecting = solving, classified by Rouché–Capelli), parallelism via directions, the tables for the plane and for space **deduced** case by case, the determinant criterion for skew lines and a complete example with a parameter. Exercises and a Going further section at the end.

2.1 The general method

Principle 5.2.1 (intersecting = solving). If two affine subspaces of $\mathbb{A}^n$ are given in implicit form, $\mathcal{F} = \{x : Ax = b\}$ and $\mathcal{G} = \{x : Cx = d\}$, their **intersection** is the solution set of the **joint system**

$$\begin{bmatrix} A \\ C \end{bmatrix} x = \begin{bmatrix} b \\ d \end{bmatrix},$$

which is classified with the **Rouché–Capelli theorem** (known in Spain as Rouché–Frobenius; Linear Algebra, Theorem 4.4.3): the intersection is empty if the ranks differ; if they coincide ($= r$), it is an affine subspace of dimension $n - r$ (Theorem 5.1.8). *If an object is given in parametric form, substitute it into the implicit equations of the other (usually the shortest route), or convert everything to implicit form (Method 5.1.9).*

Definition 5.2.2 (parallelism). Two affine subspaces are **parallel** if the direction space of one contains that of the other ($W \subseteq W'$ or $W' \subseteq W$). *(For two lines: proportional direction vectors; line and plane: the line's direction vector belongs to the plane's direction space.)*

Proposition 5.2.3. Two parallel affine subspaces are either disjoint, or one contains the other.

Proof. Suppose $W \subseteq W'$ and that $\mathcal{F} = P + W$ and $\mathcal{G} = Q + W'$ share a point R. By Proposition 5.1.6 we may take R as the base point of both: $\mathcal{F} = R + W \subseteq R + W' = \mathcal{G}$. ∎

Remark 5.2.3b (the practical test). Read backwards, the proposition tells us how to decide in practice which of the two cases occurs, and **without solving the joint system**: take **a point of the subspace with the smaller direction space** (the $\mathcal{F} = P + W$ with $W \subseteq W'$) and check whether it satisfies the equations of the other one. If it does, the two share a point and by Proposition 5.2.3 we get $\mathcal{F} \subseteq \mathcal{G}$; if it does not, they are disjoint. *(The point must be the one from the smaller object: a point of $\mathcal{G}$ decides nothing — a line contained in a plane leaves out almost every point of the plane, and seeing that one of them is not on the line says nothing about whether the line lies in the plane.)*

> Parallelism is decided **without solving anything**: it is a **rank condition on the direction vectors** (is rank(all the direction vectors together) equal to the maximum of the separate ranks?).

2.2 The tables, deduced

Each case **is deduced** in three steps: (i) set up the joint system; (ii) bound the possible ranks; (iii) read off each combination with Rouché–Capelli and the dimension count $n - r$. The tables below are the record of that deduction.

Two lines in the plane ($n = 2$)

System of 2 equations (one per line), 2 unknowns; $1 \leq \text{rank}A \leq 2$.

rank$A$	rank$[A\|b]$	Reading	Position
2	2	consistent, dim $= 0$	**intersecting** (one point)
1	1	consistent, dim $= 1$	**coincident**
1	2	inconsistent	**distinct parallel lines**

(There are no more cases: $\operatorname{rank}[A|b] \leq 2$ *and it is never smaller than* $\operatorname{rank}A$*.)*

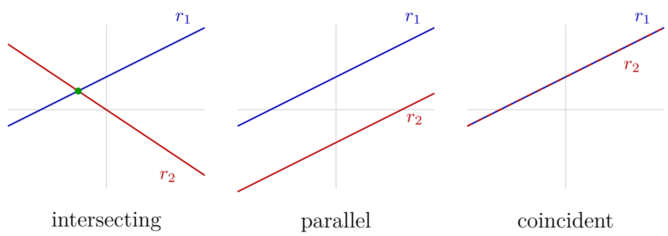

Figure 6: Relative positions of two lines in the plane (interactive version: drag the points of r_2): they meet at one point (intersecting), they do not meet (parallel) or they coincide.

[see interactive version]

In space ($n = 3$)

Two planes (2 equations): $\operatorname{rank}A \leq 2 < 3$, so if there is an intersection its dimension is ≥ 1:

Proposition 5.2.4. Two planes of $\mathbb{R}^3$ **never** meet in exactly one point.

Proof. The joint system has 2 equations and 3 unknowns: $\operatorname{rank}A \leq 2$, and if it is consistent the dimension of the intersection is $3 - \operatorname{rank}A \geq 1$. ■

$\operatorname{rank}A$	$\operatorname{rank}[A\|b]$	Position
2	2	they meet in a **line**
1	1	**coincident**
1	2	**distinct parallel planes**

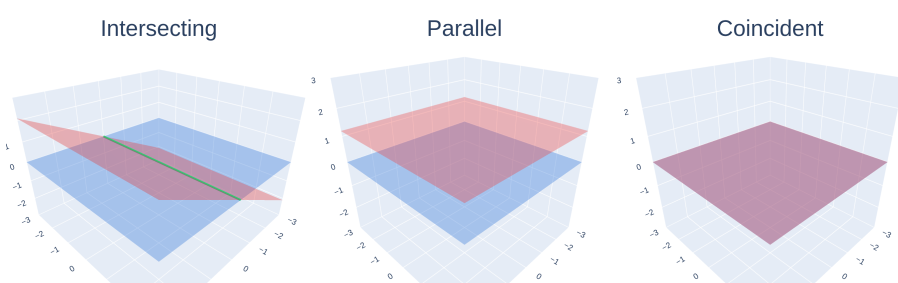

Figure 7: Relative positions of two planes in space: they meet in a line (intersecting), they are parallel or they coincide — with a slider running through the cases.

[see interactive version]

Remark 5.2.4b (the bound that makes the next two tables exhaustive). The two remaining cases involve a **line of $\mathbb{R}^3$ given in implicit form**, and its two equations are **independent by construction**: were they not, the rank would be 1 and the solution set would have dimension $3-1=2$ (Theorem 5.1.8) — a plane, not a line. Hence **each line contributes rank 2** to the joint system and $\operatorname{rank} A$ never drops below 2. With that lower bound, $\operatorname{rank} A \leq 3$ (there are only three unknowns) and $\operatorname{rank} A \leq \operatorname{rank}[A|b] \leq \operatorname{rank} A + 1$ (the augmented matrix adds a single column), the possible rank pairs are **exactly** the ones listed below: every pair with $\operatorname{rank} A = 1$ is ruled out from the start.

Line and plane (2+1 = 3 equations, 3 unknowns; $2 \leq \operatorname{rank} A \leq 3$ by Remark 5.2.4b): cases (3,3) one **point**; (2,2) the line **contained** in the plane; (2,3) **parallel** (no contact). *(The case* $\operatorname{rank} A = 3$ *inconsistent is impossible: with full rank the system is always consistent* — $\operatorname{rank}[A|b] \leq 3$.*)*

Two lines (2+2 = 4 equations, 3 unknowns; again $2 \leq \operatorname{rank} A \leq 3$): (3,3) **intersecting** at a point; (2,2) **coincident**; (2,3) **distinct parallel lines** — the pair (2,4) is impossible, because $\operatorname{rank}[A|b] \leq \operatorname{rank} A + 1$ —; and the novelty,

$$(3,4) : \textbf{skew} \text{ — they do not meet, } \textit{yet} \text{ they are not parallel.}$$

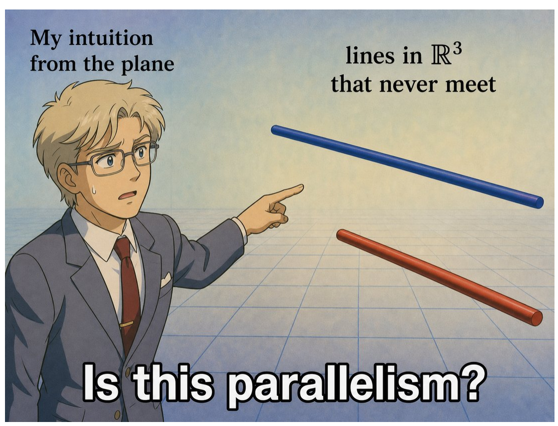

Figure 8: The plane reflex that space dismantles: in $\mathbb{R}^2$, "they do not meet" forces "parallel"; in $\mathbb{R}^3$ the new case appears — **skew lines** — and the character's question is exactly the mistake. The existence of skew lines is a dimensional phenomenon (see the note after Proposition 5.2.5).

Proposition 5.2.5 (determinant criterion for skew lines). Let $r = P + \operatorname{span}(v)$ and $s = Q + \operatorname{span}(w)$ be lines in $\mathbb{R}^3$. Then

$$r \text{ and } s \text{ are skew} \iff \det\left(\vec{PQ},\, v,\, w\right) \neq 0,$$

(the three columns: the vector connecting the lines and the two direction vectors).

Proof. ($\Leftarrow$) If the determinant is nonzero, $\{\vec{PQ}, v, w\}$ is independent (Linear Algebra, Theorem 4.6.11). In particular v, w are not proportional (not parallel) and the lines do not meet: if they shared a point R, then $\vec{PQ} = \vec{PR} + \vec{RQ}$ would be a combination of v and w (since $\vec{PR} \parallel v$, $\vec{RQ} \parallel w$), against the independence. ($\Rightarrow$) Contrapositive: if the determinant is 0, the family is dependent. If v, w are proportional, the lines are parallel (not skew). If not, $\{v, w\}$ is independent and then $\vec{PQ} \in \operatorname{span}(v, w)$ (in a nontrivial dependence, the coefficient of $\vec{PQ}$ cannot be 0, and one solves for it — field). Writing $\vec{PQ} = \alpha v + \beta w$, the point $P + \alpha v = Q - \beta w$ lies on both lines: they meet (not skew). ■

> **Why there are no skew lines in the plane:** the criterion would require three independent vectors in K^2, impossible (Linear Algebra, exercise 8 of Session 3: the dimension forbids it). The existence of skew lines is a **dimensional** phenomenon: room is needed.

> **Example 5.2.6 (a skew pair).** $r : (0,0,0) + t(1,0,0)$ (the x-axis) and $s : (0,0,1) + t(0,1,0)$. With $\vec{PQ} = (0,0,1)$: $\det \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} = 1 \neq 0$ — skew. *(Intuition: two corridors on different floors with perpendicular directions.)*

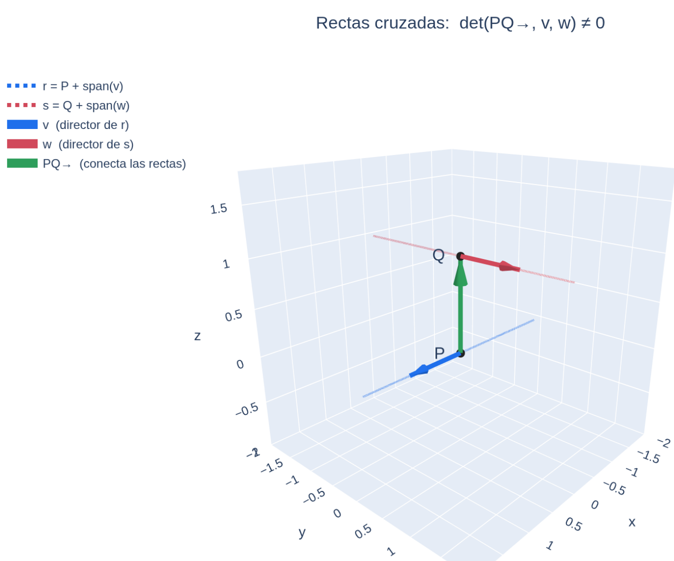

Figure 9: The three columns of the determinant in Proposition 5.2.5, on the skew pair of Example 5.2.6: the direction vectors v and w issuing from P and Q, and the vector $\vec{PQ}$ joining the two lines. That the three are not coplanar **is** the determinant being nonzero; rotating the figure shows that r and s do not meet, even though head-on they seem to.

[see interactive version]

2.3 Example with a parameter (complete)

Example 5.2.7 (the three regimes). Position of $r : (x, y, z) = (0, 0, 1) + t\,(1,\ a^2 - 2,\ 0)$ and $\pi : x + y = a - 1$, according to $a \in \mathbb{R}$.

Substituting the points of the line, $(t,\ (a^2 - 2)t,\ 1)$, into the implicit equation:

$$t + (a^2 - 2)\,t = a - 1 \iff (a^2 - 1)\,t = a - 1.$$

A single equation in t, and its analysis is the trichotomy of Session 1 of Linear Algebra in geometric guise:

- $a \neq \pm 1$**:** nonzero coefficient $\to$ unique solution $t = \frac{a-1}{a^2-1} = \frac{1}{a+1}$: the line **meets** the plane at one point.
- $a = 1$**:** it becomes $0 \cdot t = 0$, which **always** holds: the line is **contained** in the plane. *(Check: direction* $(1, -1, 0)$ *and plane* $x + y = 0$*; the points* $(t, -t, 1)$ *all satisfy it.)*
- $a = -1$**:** it becomes $0 \cdot t = -2$, which **never** holds: line and plane **parallel** (no contact).

  Note the mechanics: the regime is decided by **two different things** — whether the coefficient vanishes (parallelism of direction) and, in that case, whether or not the constant term vanishes (contained vs. strictly parallel). That is why the parameter had to appear on both sides, with zeros at different values of a.

Remark 5.2.8 (where the coefficient and the constant term come from). The computation of the example is general, and it is worth writing it down. For the line $r : X = P + t\,v$ and the plane $\pi : ax_1 + bx_2 + cx_3 = d$, substituting the points of r into the equation of π leaves **a single equation in** t, $\alpha\,t = \beta$, with

$$\alpha = av_1 + bv_2 + cv_3, \qquad \beta = d - (aP_1 + bP_2 + cP_3).$$

That is: α is what comes out of feeding the **direction vector** into the left-hand side of the plane, and β measures how far the **base point** is from satisfying the equation. Hence the trichotomy, now justified term by term:

- $\alpha \neq 0$: unique solution $t = \beta/\alpha$ — the line **meets** the plane at one point.
- $\alpha = 0$: since the directions of π are the solutions of the **homogeneous** equation $ax_1 + bx_2 + cx_3 = 0$ (Theorem 5.1.8), $\alpha = 0$ is exactly the statement that v is a direction of π, that is, that there is **parallelism** (Definition 5.2.2). And then β decides: with $\beta = 0$ the equation always holds and the line is **contained**; with $\beta \neq 0$ it never holds and they are **strictly parallel**.

(In the language of Session 3, $\alpha = n \cdot v$ *with* $n = (a, b, c)$ *the normal of the plane: parallelism is orthogonality between the direction vector and the normal.)*

Exercises

1. ★ Analyze the relative position, setting up the system and comparing ranks:
   (a) the lines of $\mathbb{R}^2$: $x + y = 1$ and $2x + 2y = 3$;
   (b) the planes $x - y + z = 0$ and $2x - 2y + 2z = 5$;
   (c) the lines of $\mathbb{R}^3$: $r : (1, 0, 0) + t(1, 1, 0)$ and $s : (0, 1, 0) + t(0, 1, 1)$ — also apply the determinant criterion (Proposition 5.2.5).
2. ★ Find the intersection of $\pi_1 : x+y+z = 3$ and $\pi_2 : x-y = 1$, parametrized, and check that its dimension agrees with $3 - \text{rank}$.
3. ★ Analyze according to a: the line $r : (x, y, z) = (0, 0, 1) + t(1, a, 0)$ and the plane $\pi : x + y = 2$. **Only two** of the three possible regimes appear — identify them, give the critical value of a, and **explain why the case "line contained in the plane" is impossible for every** a *(hint: what two things would have to vanish at once, and can they?).*
4. ★ Prove that the lines $r : (0, 0, 0) + t(1, 1, 0)$ and $s : (1, 0, 1) + t(1, 1, 1)$ are skew, and give implicit equations for each.
5. Deduce the line–plane table in $\mathbb{R}^3$: list the possible rank pairs of the joint system and justify why (3, inconsistent) does not appear.
6. Prove: if two lines of $\mathbb{R}^3$ meet at **two** distinct points, they are coincident. *(Hint: two points determine the direction.)*
7. **(Conceptual.)** Explain why Definition 5.2.2 of parallelism includes the case "one contained in the other", and why that is consistent with Proposition 5.2.3.

Going further and references

Going further (pencils of planes). The planes containing a given line $r = \pi_1 \cap \pi_2$ are exactly those with equation $\lambda(\text{eq. of } \pi_1) + \mu(\text{eq. of } \pi_2) = 0$ with $(\lambda, \mu) \neq (0, 0)$: the **pencil of planes** with axis r. Useful for imposing "a plane containing r and satisfying X" without parametrizing. See de Burgos.

References. Relative positions via ranks: de Burgos, *Álgebra lineal y geometría cartesiana*; Hernández, *Álgebra y geometría.* Skew lines and the determinant criterion: de Burgos. The underlying analysis of systems: the Linear Algebra notes, Session 4.

Lecture Notes · Geometry · Session 3 — Euclidean space: dot product, distances and angles

The full metric structure: the dot product and its properties proved, Cauchy–Schwarz with proof (and from it the well-defined angle and the triangle inequality), Pythagoras, the orthogonal projection derived and the distance formulas worked out. The counterfactual of why this structure demands $\mathbb{R}$, resolved in detail. Exercises and a Going further section at the end.

3.1 The dot product

Until now the geometry was **affine**: incidence, parallelism, dimensions. To speak of lengths and angles an additional structure is needed, and a single operation brings it.

Definition 5.3.1 (inner product). Let V be a vector space **over** $\mathbb{R}$. An **inner product** on V is a map

$$\langle\,\cdot\,,\,\cdot\,\rangle \;:\; V \times V \longrightarrow \mathbb{R}, \qquad (u, v) \longmapsto \langle u, v\rangle$$

—two vectors in, one **real** number out— such that, for all $u, v, w \in V$ and $a, b \in \mathbb{R}$:

1. **Symmetry:** $\langle u, v\rangle = \langle v, u\rangle$.
2. **Bilinearity:** $\langle au + bw,\, v\rangle = a\langle u, v\rangle + b\langle w, v\rangle$ (and, by symmetry, likewise in the second argument).
3. **Positive definiteness:** $\langle u, u\rangle \geq 0$, with equality **if and only if** $u = 0$.

   Where exactly $\mathbb{R}$ is needed. Not in V as a set of vectors —all that is asked of V is that it be a real vector space— but in the **codomain**: axiom 3 compares the result with 0, and for that the target field must be **ordered**. That, and not before, is where this geometry stops working over an arbitrary field.

Proposition 5.3.2 (the standard product on $\mathbb{R}^n$ is an inner product). On $V = \mathbb{R}^n$, the formula

$$u \cdot v = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$$

satisfies the three axioms. It is the one we use throughout the module unless stated otherwise. 1. **Symmetry:** $u \cdot v = v \cdot u$. 2. **Bilinearity:** $(au + bw) \cdot v = a(u \cdot v) + b(w \cdot v)$. 3. **Positivity:** $u \cdot u \geq 0$, with equality **if and only if** $u = 0$.

Proof. (1) and (2): straight from the definition (commutativity and distributivity of $\mathbb{R}$, term by term). (3) $u \cdot u = u_1^2 + \cdots + u_n^2$ is a sum of real squares, ≥ 0; and a sum of real squares is 0 only if **each** summand is 0, that is, $u = 0$. ■

Convention for the rest of the session. Everything that follows —Cauchy–Schwarz, norm, distance, angle, orthogonality and projection— is stated and proved using **only the three axioms**, so it holds in any real vector space V with an inner product, not just in $\mathbb{R}^n$. We write $\mathbb{R}^n$ with its standard product when **coordinates** are needed —in examples and computations— and V when the statement does not need them. *(Same criterion as in Linear Algebra with "if K is a field…".)*

Remark 5.3.3 (where $\mathbb{R}$ is used — the thread of fields, final twist). Property (3) is the one that does **not survive** in an arbitrary field:

- In $K = \mathbb{Z}/5\mathbb{Z}$, with $u = (1, 2)$: $u{\cdot}u = 1{+}4 = 5 = 0$, with $u \neq 0$. "Positivity" cannot even be stated ($\mathbb{Z}/5\mathbb{Z}$ is not ordered) and its consequence fails outright.
- In $\mathbb{C}$ with the same formula it is even starker: $u = (0, i) \neq 0$ gives $u \cdot u = 0 + i^2 = -1$, a **negative** "squared length"; and $u = (1, i)$ gives $u \cdot u = 0$ with $u \neq 0$. *(That is why in $\mathbb{C}$ a different operation is used, with conjugates — see Going further.)*
- Moreover, the **norm** (below) needs square roots of nonnegative numbers, which $\mathbb{R}$ guarantees.

Conclusion: **affine** geometry works over any field K (Sessions 1–2); **Euclidean** geometry — this session — lives over $\mathbb{R}$.

3.2 Norm, distance and the key inequality

Definition 5.3.4. The **norm** of u: $\|u\| = \sqrt{u \cdot u}$ (well defined by 5.3.2.3). The **distance** between two points: $d(P, Q) = \|\vec{PQ}\|$. *(In the plane, $\|(a, b)\| = \sqrt{a^2 + b^2}$: the Pythagorean theorem turned into a definition.)*

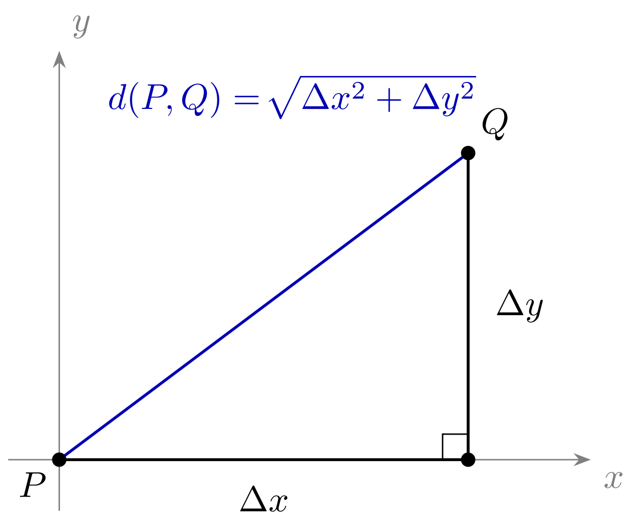

Figure 10: Distance in the plane via Pythagoras: $d(P, Q) = \sqrt{\Delta x^2 + \Delta y^2}$, the hypotenuse of the triangle with legs $\Delta x, \Delta y$.

Immediate properties: $\|u\| \geq 0$ with equality only at $u = 0$; $\|\lambda u\| = |\lambda|\|u\|$ (a λ^2 comes out of the bilinearity and the square root returns it as $|\lambda|$).

Lemma 5.3.4b (expanding the square). For u, w in V and $t \in \mathbb{R}$:

$$\|u + w\|^2 = \|u\|^2 + 2\langle u, w\rangle + \|w\|^2,$$

$$\|u - tw\|^2 = \|u\|^2 - 2t\,\langle u, w\rangle + t^2\|w\|^2.$$

Proof. Expand $\langle u + w,\, u + w\rangle$ by bilinearity and merge the two cross terms into one, which is where symmetry is used; the second identity is the first with $w \rightsquigarrow -tw$. ■

> One line of algebra, stated separately because it **gets used three times**: in the proof of Cauchy–Schwarz, in Pythagoras, and when deriving the orthogonal projection. It is the analogue of $(a \pm b)^2 = a^2 \pm 2ab + b^2$, with the inner product playing the role of the product.

Theorem 5.3.5 (Cauchy–Schwarz inequality). Let V be real with an inner product. For all $u, v \in V$:

$$|u \cdot v| \leq \|u\|\,\|v\|,$$

with equality **if and only if** u and v are proportional.

Proof. If $v = 0$, both sides are 0 (and $0 = 0 \cdot u$: proportional). Let $v \neq 0$. For every $t \in \mathbb{R}$, positivity gives

$$0 \leq \|u - tv\|^2 = \|u\|^2 - 2t\,(u \cdot v) + t^2\|v\|^2,$$

(using bilinearity and symmetry). The right-hand side is a quadratic polynomial in t with leading coefficient $\|v\|^2 > 0$ that is never negative: its **discriminant** is ≤ 0:

$$4(u \cdot v)^2 - 4\|v\|^2\|u\|^2 \leq 0 \iff |u \cdot v| \leq \|u\|\|v\|.$$

Equality: the discriminant is 0 exactly when the polynomial has a root t_0, that is, $\|u - t_0 v\| = 0$, that is (Proposition 5.3.2.3), $u = t_0 v$: proportional. Conversely, if $u = t_0 v$ the equality is checked directly. ■

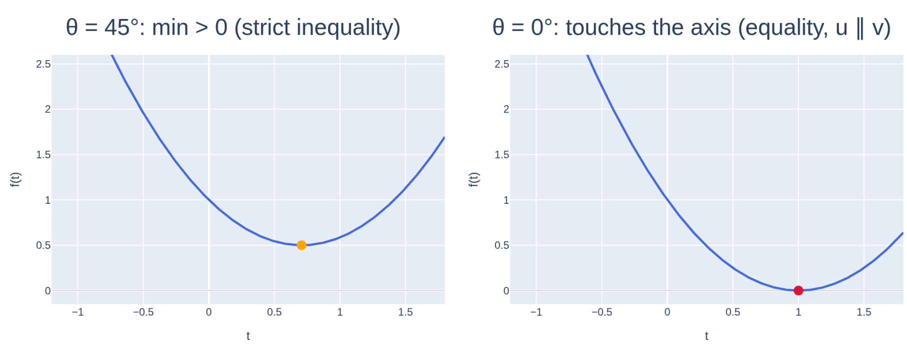

Figure 11: Cauchy–Schwarz, made visible: $f(t) = \|u - tv\|^2$ is a parabola that never dips below the axis; its discriminant being ≤ 0 is the inequality, and it touches the axis (equality) only if u, v are proportional.

[see interactive version]

Corollary 5.3.6 (triangle inequality). $\|u + v\| \leq \|u\| + \|v\|$.

Proof. Expanding and bounding the cross term,

$$\begin{aligned}\|u+v\|^2 &= \|u\|^2 + 2(u\cdot v) + \|v\|^2 \\ &\le \|u\|^2 + 2\|u\|\|v\| + \|v\|^2 = (\|u\| + \|v\|)^2,\end{aligned}$$

using $u \cdot v \le |u \cdot v| \le \|u\|\|v\|$ (Cauchy–Schwarz); conclude by taking square roots (both sides are ≥ 0). ■

3.3 Angle and orthogonality

The goal is to **measure the angle** between two vectors. The starting point is an identity in the plane which is **neither a definition nor something read off a picture**: it has to be proved.

Proposition 5.3.6b (the cosine formula, in $\mathbb{R}^2$). If $u, v \in \mathbb{R}^2$ are nonzero and θ is the angle between them, then

$$u \cdot v = \|u\|\,\|v\| \cos\theta.$$

Proof. Write both in polar form: $u = \|u\|(\cos\alpha, \sin\alpha)$ and $v = \|v\|(\cos\beta, \sin\beta)$, with α, β their arguments. Then

$$u \cdot v = \|u\|\,\|v\|\,(\cos\alpha\cos\beta + \sin\alpha\sin\beta) = \|u\|\,\|v\|\cos(\alpha - \beta),$$

by the cosine-of-a-difference formula (Complex numbers module, §2.2), and $\alpha - \beta$ is —up to sign and up to 2π, both harmless since the cosine is even and periodic— the angle θ between u and v. ■

*(Second route, **and one must pick one**: taking the **law of cosines** as prior classical geometry, applied to the triangle with sides u, v and $u - v$, it gives $\|u - v\|^2 = \|u\|^2 + \|v\|^2 - 2\|u\|\|v\|\cos\theta$; expanding by bilinearity, $\|u - v\|^2 = \|u\|^2 - 2(u \cdot v) + \|v\|^2$, and comparing yields the same identity. **We follow the first**, via polar form, because exercise 7 goes the other way —it derives the law of cosines from this identity— and using both at once would be circular.)*

> **And in dimension n? It holds just the same, with no new computation.** Two nonzero vectors span a subspace $W = \operatorname{span}(u, v)$ of dimension ≤ 2: the whole geometry of u and v takes place **inside that plane**. Moreover the three quantities involved —$u \cdot v$, $\|u\|$, $\|v\|$— depend only on u and v, not on the ambient space, so they may be computed inside W. Taking in W the orthonormal basis
>
> $$e_1 = \frac{u}{\|u\|}, \qquad e_2 = \frac{v - (v \cdot e_1)\, e_1}{\|v - (v \cdot e_1)\, e_1\|}$$
>
> —the second one is exactly v minus its projection onto u, normalized (§3.4)—, coordinates identify W with $\mathbb{R}^2$ **preserving inner products and norms**, and the proposition applies there. If u and v

are proportional, $\dim W = 1$, the denominator of e_2 vanishes, and the case is checked directly: $\theta = 0$ or π, and the formula gives $u \cdot v = \pm\|u\|\|v\|$.

The upshot is what matters: Definition 5.3.7 does **not invent** an angle for $\mathbb{R}^n$ — it returns exactly the usual Euclidean angle, measured inside the plane the two vectors span.

What Proposition 5.3.6b does **not** provide is an angle in $\mathbb{R}^n$: there, none is defined beforehand. So the statement is turned around and the equality is **taken as the definition** —consistent with the plane, by what we have just proved—, and Cauchy–Schwarz guarantees that it makes sense.

Definition 5.3.7. For $u, v \neq 0$, the **angle** $\theta \in [0, \pi]$ between them is the unique one with

$$\cos\theta = \frac{u \cdot v}{\|u\|\,\|v\|}.$$

Why it is well defined (the point usually omitted): by Cauchy–Schwarz, the quotient lies in $[-1, 1]$, which is exactly the range of the cosine on $[0, \pi]$, where the cosine is injective. Without Theorem 5.3.5, the definition would not even make sense. Rewritten: $u \cdot v = \|u\|\|v\|\cos\theta$ — the dot product measures "how much u points in the direction of v" (positive: acute; zero: right; negative: obtuse).

The shadow has a sign. The visual reading of the formula is that $\|u\|\cos\theta$ is the *shadow* of u on the direction of v. With θ **acute** — the case in the picture — that shadow is the adjacent leg of a right triangle with hypotenuse u, and it is a length. With θ **obtuse** the cosine is negative and the shadow falls to the other side: there is no leg left to measure, but the formula still holds unchanged, because what $\|u\|\cos\theta$ measures is a **signed length** (positive if the shadow points along v, negative if it points the other way). It is the same information as the sign of the dot product.

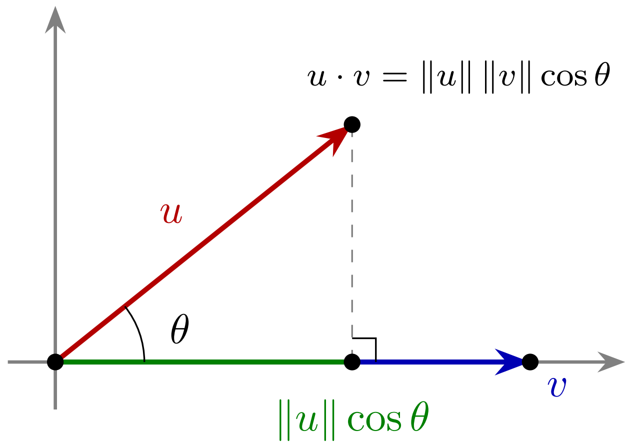

Figure 12: The dot product as a projection: $u \cdot v = \|u\|\,\|v\|\cos\theta$, with $\|u\|\cos\theta$ the shadow of u on v.

Definition 5.3.8. $u \perp v$ (**orthogonal**) if $u \cdot v = 0$.

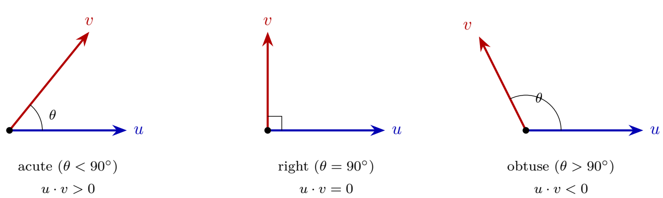

Figure 13: The sign of $u \cdot v$ according to the angle: acute (> 0), right ($= 0$, orthogonal), obtuse (< 0).

Corollary 5.3.9.1 (Pythagoras). If $u \perp v$, then $\|u + v\|^2 = \|u\|^2 + \|v\|^2$.

Proof. By Lemma 5.3.4b, $\|u + v\|^2 = \|u\|^2 + 2\langle u, v\rangle + \|v\|^2$, and $\langle u, v\rangle = 0$ by hypothesis. ■ *(It is literally Pythagoras' theorem: with perpendicular legs, the square of the hypotenuse is the sum of the squares.)*

Proposition 5.3.9.2. For fixed $v \neq 0$, the set $\{v\}^\perp = \{u \in V : \langle u, v\rangle = 0\}$ is a **subspace** of V (by bilinearity). In $V = \mathbb{R}^n$ it has dimension $n - 1$: a vector hyperplane.

Proof. It is the kernel of the row matrix v^T, of rank 1 since $v \neq 0$; by rank–nullity (Linear Algebra) its dimension is $n - 1$. ■

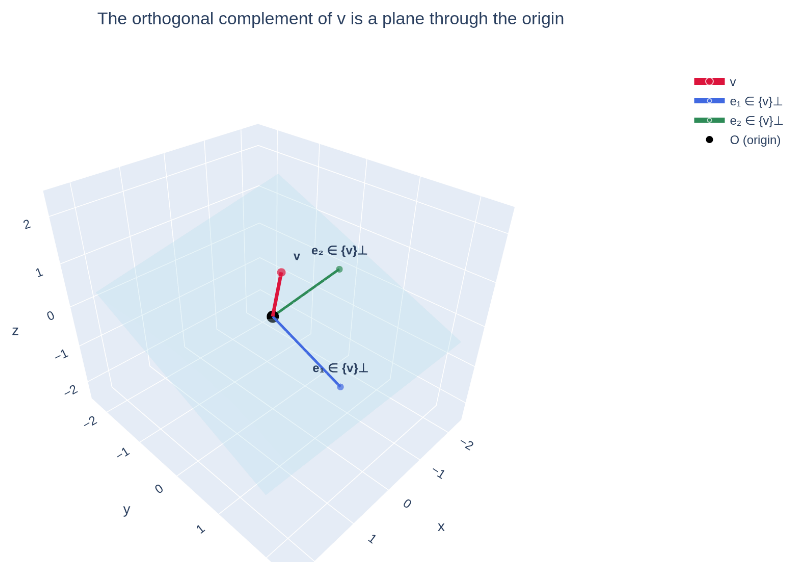

Figure 14: The orthogonal complement of a nonzero vector v is a plane through the origin: $\{v\}^\perp = \{u : u \cdot v = 0\}$. It is the basis of the idea of the “normal” of a plane.

[see interactive version]

3.4 Orthogonal projection and distances

Proposition 5.3.10 (orthogonal projection, derived). Given u and $v \neq 0$, there exists a unique multiple of v, called the **orthogonal projection of** u **onto** v, such that the remainder is perpendicular to v; and it equals

$$\operatorname{proj}_v(u) = \frac{u \cdot v}{\|v\|^2}\, v.$$

Proof. We look for t with $(u - tv) \perp v$:

$$(u - tv) \cdot v = 0 \iff u \cdot v - t\,\|v\|^2 = 0 \iff t = \frac{u \cdot v}{\|v\|^2},$$

(solvable because $\|v\|^2 \neq 0$). The condition determines t uniquely, so the projection is unique. ∎

> **In what sense it is unique** (the usual confusion). It is not claimed that there is only one vector perpendicular to v —there is a whole subspace, $\{v\}^\perp$— but that **among the multiples of** v **there is exactly one** whose remainder comes out perpendicular to v. As a decomposition: every u is written **in one and only one way** as
>
> $$u = \underbrace{\operatorname{proj}_v(u)}_{\text{multiple of } v} + \underbrace{(u - \operatorname{proj}_v(u))}_{\perp\, v},$$

and the fact that both pieces are described "with respect to v" is no contradiction: one runs **along** v and the other **perpendicular** to it. *(Uniqueness is inherited from the system: the condition is a linear equation in t with coefficient $\|v\|^2 \neq 0$.)*

The formula **falls out of the geometric condition in one line**. The resulting decomposition $u = \mathrm{proj}_v(u) + (u - \mathrm{proj}_v(u))$ splits u into "part parallel to v" + "part perpendicular".

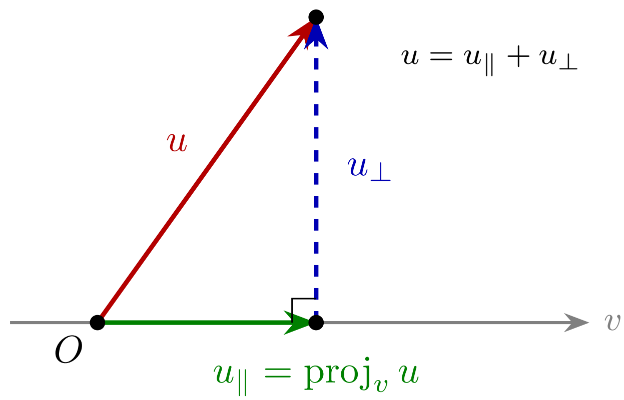

Figure 15: Orthogonal decomposition: $u = u_\parallel + u_\perp$, with $u_\parallel = \mathrm{proj}_v u$ the component along v and $u_\perp$ the perpendicular one.

Method 5.3.11 (distance from a point to a line). For Q and the line $r = P + \mathrm{span}(v)$: decompose $\vec{PQ}$ with the projection onto v; the distance is the norm of the **perpendicular part**:

$$d(Q, r) = \left\|\vec{PQ} - \mathrm{proj}_v(\vec{PQ})\right\|.$$

(That this really is the minimum of $d(Q, X)$ over $X \in r$ follows from Pythagoras: for any other point of the line, the squared distance adds the perpendicular part — fixed — plus a nonzero parallel part. Details: exercise 6.)

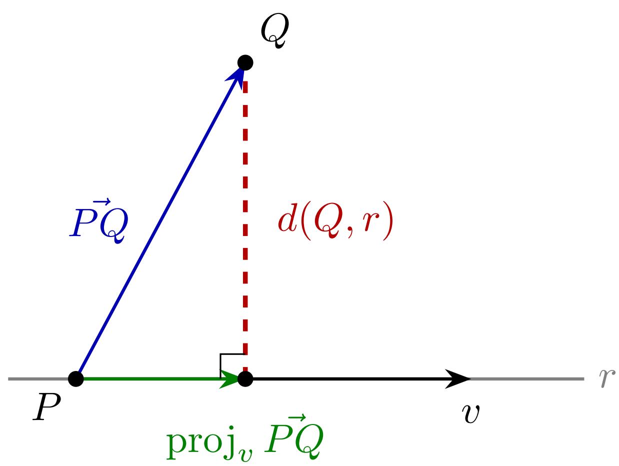

Figure 16: Distance from Q to the line r: the length of the component of $\vec{PQ}$ perpendicular to the direction v.

Proposition 5.3.12 (the normal of a plane, proved). In $\mathbb{R}^3$, the vector $n = (a, b, c)$ is **orthogonal to all the directions** of the plane $\pi : ax+by+cz = d$.

Proof. The equation says exactly $n \cdot X = d$ for the points X of the plane (dot product against the coordinate vector). If $X, Y \in \pi$, subtracting: $n \cdot \vec{XY} = n \cdot Y - n \cdot X = d - d = 0$. Every direction of the plane is of the form $\vec{XY}$: orthogonal to n. ■

> **The symmetry hidden in codimension** (why a line and a hyperplane are given by the same data). A **line** is determined by a point and **one** vector, its direction; and a **hyperplane** too, by a point and **one** vector, its normal. It looks like a coincidence, and it is exactly codimension: the line has dimension 1 and one gives it *the dimension it has* —its direction—; the hyperplane has dimension $n-1$, and giving it $n-1$ directions would be absurd, so one gives it **the only one it lacks**, the one not living in it. In one case we describe what the object **has**; in the other, what the ambient space has **left over**.
>
> And the dictionary with implicit equations (§1.2) falls out: substituting the point into the equation gives the constant term, $d = \langle n, P\rangle$, and the normal vector **is** the list of coefficients. Implicit equations of a hyperplane were never a separate notation: they are a point and a direction, written differently.

Method 5.3.13 (distance from a point to a plane). For Q and $\pi : ax + by + cz = d$ with normal n: take any $P \in \pi$ and project $\vec{PQ}$ onto n — the part parallel to the normal is what separates Q from the plane:

$$d(Q,\pi) = \left\|\mathrm{proj}_n(\vec{PQ})\right\| = \frac{|n\cdot \vec{PQ}|}{\|n\|} = \frac{|a\,q_1 + b\,q_2 + c\,q_3 - d|}{\sqrt{a^2+b^2+c^2}},$$

where the last equality uses $n \cdot \vec{PQ} = n \cdot Q - n \cdot P = (aq_1 + bq_2 + cq_3) - d$ (Proposition 5.3.12). The point–plane distance formula is thus **derived** in two lines: projection + the reading of the plane's equation.

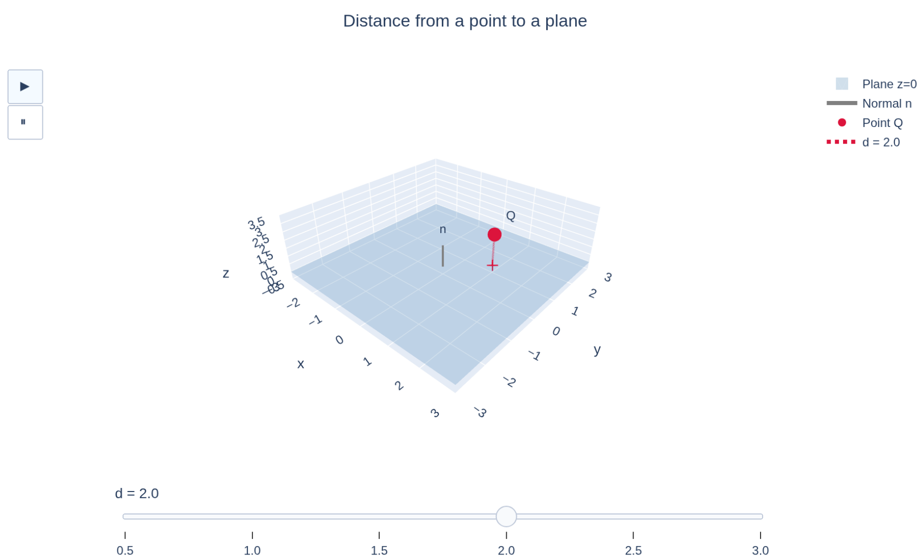

Figure 17: Distance from a point Q to a plane: the length of the perpendicular segment from Q to the plane (its projection onto the normal n).
[see interactive version]

Definition 5.3.14 (angle between planes). The **angle between two intersecting planes** is the **acute** angle between their normals:

$$\cos\theta = \frac{|n_1 \cdot n_2|}{\|n_1\|\,\|n_2\|}, \qquad \theta \in [0, \tfrac{\pi}{2}].$$

The absolute value is not decoration: the normal of a plane is defined only up to sign (n and $-n$ describe the same plane), and the angle between the planes cannot depend on that choice — the absolute value makes it irrelevant. (That the angle "between the walls" agrees with the angle between the normals is a quarter-turn of the picture: each normal is perpendicular to its plane.)

Exercises

1. ★ For $u = (1, 2, 2)$ and $v = (2, -1, 0)$: compute $u \cdot v$, $\|u\|$, $\|v\|$, the angle between them, and decide whether they are orthogonal. Verify Cauchy–Schwarz numerically.
2. ★ Project $u = (3, 1)$ onto $v = (1, 1)$, draw the decomposition (parallel + perpendicular) and compute the distance from the point $(3, 1)$ to the line $\operatorname{span}((1, 1))$.
3. ★ Compute the distance from the point $Q = (1, 1, 1)$ to the plane $2x - y + 2z = 1$ following Method 5.3.13 step by step (choose the point P of the plane yourself), and check with the final formula.
4. ★ **(Structural counterfactual.)** In $K = \mathbb{Z}/5\mathbb{Z}$, check that $u = (1, 2)$ satisfies $u \cdot u = 0$ with $u \neq 0$. Explain: (a) which property of Proposition 5.3.2 fails; (b) why it cannot even be *stated* in $\mathbb{Z}/5\mathbb{Z}$ (what is the field missing?); (c) which geometric notion becomes impossible.
5. Prove that $\{v\}^\perp$ is a subspace using only bilinearity (no matrices), and compare with the proof of 5.3.9.2.
6. Complete the details of Method 5.3.11: if $X = P + sv \in r$, prove with Pythagoras that $d(Q, X)^2 = d(Q, r)^2 + \|t_0 v - sv\|^2$ for a certain t_0, and conclude that the minimum is attained at the projection.
7. Prove the **law of cosines**: $\|u - v\|^2 = \|u\|^2 + \|v\|^2 - 2\|u\|\|v\| \cos\theta$. *(Expand $\|u - v\|^2$ by bilinearity and use Definition 5.3.7. Note the direction: here the law of cosines* ***is derived*** *from the definition of angle — which is why Proposition 5.3.6b is proved via polar form and not from it.)*
8. True or false? “If $u \cdot v = u \cdot w$ with $u \neq 0$, then $v = w$.” Justify or give a counterexample (in $\mathbb{R}^3$!).
9. ★ **(Gram–Schmidt, step by step.)** In $\mathbb{R}^3$, with $v_1 = (1, 1, 0)$ and $v_2 = (1, 0, 1)$:
   (a) Normalize the first: check that $e_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$ is a unit vector.
   (b) Compute $\langle v_2, e_1 \rangle$ and subtract from v_2 its projection onto e_1: you should get $w_2 = (\frac{1}{2}, -\frac{1}{2}, 1)$.
   (c) Check that $w_2 \perp e_1$ — **that is the whole point of the method**: subtracting the projection removes exactly the part visible from e_1.
   (d) Normalize: $e_2 = \frac{1}{\sqrt{6}}(1, -1, 2)$. Verify $\|e_2\| = 1$.
   (e) Check that $\operatorname{span}(e_1, e_2) = \operatorname{span}(v_1, v_2)$, which is what makes the result a **basis** of the same plane.
   (f) With the orthonormal basis in hand, compute $\langle v_1, v_2 \rangle$ in two ways —directly, and by summing products of coordinates in the basis $\{e_1, e_2\}$— and check they agree.

Going further and references

Going further (general inner products). The three properties of Proposition 5.3.2 are taken as **axioms** and define the *inner product spaces*, where this whole section reproduces word for word (Cauchy–Schwarz included: the proof of 5.3.5 only used the axioms

— check it). In $\mathbb{C}^n$ the right product is $\sum u_i\overline{v_i}$ (with conjugates, to rescue positivity). See Axler, *Linear Algebra Done Right*, ch. 6.

Going further (Gram–Schmidt: manufacturing orthonormal bases). A basis $\{v_1, \dots, v_k\}$ is **orthonormal** if its vectors are unit and pairwise orthogonal; in such a basis coordinates are read off with inner products —$v = \sum\langle v, e_i\rangle e_i$— and the inner product is computed by summing products of coordinates, as in $\mathbb{R}^k$. Every basis can be turned into an orthonormal one by **subtracting projections**, which is all the **Gram–Schmidt** process does: set $w_1 = v_1$ and, at each step, $w_j = v_j - \sum_{i<j} \operatorname{proj}_{w_i}(v_j)$ —the vector v_j minus whatever the previous ones already cover (§3.4)—, normalizing at the end, $e_i = w_i/\|w_i\|$. The span never changes, and no w_j comes out zero if the v_j were independent. It is the general case of the two-step trick used in the remark after Proposition 5.3.6b, and exercise 9 walks through it with numbers. See Hefferon, *Linear Algebra*, Three.VI, or Axler, ch. 6.

Going further (least squares). The orthogonal projection onto a subspace (not just a line) solves the problem of "the best approximate solution" of an inconsistent system — the **least squares** regression of data science. Strang, §4.2–4.3.

Optional going further (distances in general — metric spaces). The Euclidean distance is one case of a broader concept. A **distance** (or *metric*) on a set X is a function $d : X \times X \to \mathbb{R}_{\geq 0}$ with: (i) $d(x, y) = 0 \iff x = y$; (ii) **symmetry**, $d(x, y) = d(y, x)$; (iii) the **triangle inequality**, $d(x, z) \leq d(x, y) + d(y, z)$. The Euclidean one satisfies them (the triangle inequality is Corollary 5.3.6). There are other useful ones: the taxicab distance $\sum_i |x_i - y_i|$ or the maximum distance $\max_i |x_i - y_i|$. Further reading: **W. A. Sutherland, *Introduction to Metric and Topological Spaces* (Oxford University Press)**, ch. 2; it will reappear in later Analysis/Topology courses. Not assessed.

References. Dot product and Euclidean geometry: Hernández, *Álgebra y geometría*; de Burgos. Cauchy–Schwarz and inner products: Axler, ch. 6.A. Projections: Strang, §4.2.

Lecture Notes · Geometry · Session 4 — Cross product, areas and volumes

The closing session of the module, with everything proved: the properties of the cross product (including Lagrange's identity and from it the interpretation as an area), the scalar triple product identified with the determinant, the coplanarity tests, and the distance between skew lines derived. The "symbolic determinant" mnemonic justified. Exercises and a Going further section at the end.

4.1 The cross product in $\mathbb{R}^3$

The geometry of space constantly calls for a vector **perpendicular to two others** (normals of planes, above all). The cross product constructs it explicitly.

Definition 5.4.1. For $u = (u_1, u_2, u_3)$, $v = (v_1, v_2, v_3) \in \mathbb{R}^3$:

$$u \times v = (u_2v_3 - u_3v_2,\ \ u_3v_1 - u_1v_3,\ \ u_1v_2 - u_2v_1) \in \mathbb{R}^3.$$

Remark 5.4.2 (the mnemonic, justified). The "symbolic determinant" rule

$$u \times v = \text{"}\det\text{"} \begin{pmatrix} e_1 & e_2 & e_3 \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{pmatrix}$$

is not a definition (it mixes vectors and scalars in one matrix), but it works as a computation rule: expanding formally along the first row (cofactors, Sarrus' rule by blocks), the three "coefficients" of e_1, e_2, e_3 that come out are exactly the three components of Definition 5.4.1 — check it (exercise 7). Its real merit appears in Proposition 5.4.6.

Proposition 5.4.3 (algebraic properties). 1. **Antisymmetry:** $v \times u = -(u \times v)$; in particular $u \times u = 0$. 2. **Bilinearity:** linear in each argument. 3. $u \times v = 0 \iff u, v$ linearly dependent (proportional).

Proof. (1) and (2): direct componentwise check (each component is a bilinear antisymmetric expression in the pairs of coordinates). (3) ($\Leftarrow$) If $v = \lambda u$, each component is of the type $u_i(\lambda u_j) - u_j(\lambda u_i) = 0$ (or $u = 0$, trivial). ($\Rightarrow$) Let M be the 2×3 matrix with rows u, v. If they are independent, $\operatorname{rank} M = 2$: reducing to echelon form leaves **two pivots**, in certain columns $j < k$. Consider the 2×2 submatrix of columns j, k — a **minor**. Row operations on M act **on entire rows**, so they induce exactly row operations on that submatrix; in the echelon form, the submatrix of the pivot columns is triangular with nonzero diagonal, that is, invertible — and since row operations preserve invertibility (they are multiplication by elementary matrices, Linear Algebra 4.2.15.2 and 4.3.24), the **original** submatrix is invertible: its determinant, nonzero (Linear Algebra, Theorem 4.6.8). But the three 2×2 minors of M are, up to sign, **the three components of** $u \times v$ (compare with Definition 5.4.1): some component is nonzero and $u \times v \neq 0$. ■

Proposition 5.4.4 (perpendicularity). $(u \times v) \perp u$ and $(u \times v) \perp v$.

Proof. Direct:

$$(u \times v) \cdot u = (u_2v_3 - u_3v_2)u_1 + (u_3v_1 - u_1v_3)u_2 + (u_1v_2 - u_2v_1)u_3 = 0,$$

because the six terms cancel in pairs ($u_1u_2v_3$ with $-u_2u_1v_3$, etc.). Likewise with v. ■

Theorem 5.4.5 (Lagrange's identity and the area). For arbitrary $u, v \in \mathbb{R}^3$,

$$\|u \times v\|^2 = \|u\|^2\|v\|^2 - (u \cdot v)^2,$$

and therefore, **if** $u, v \neq 0$ (which is what Definition 5.3.7 requires for θ to be defined),

$$\|u \times v\| = \|u\|\,\|v\|\,\sin\theta,$$

which is the **area of the parallelogram** with sides u and v.

Proof. The identity is a computation: expanding $\|u\times v\|^2$ (the sum of the squares of the three components) and $\|u\|^2\|v\|^2 - (u\cdot v)^2$ (products of sums), both sides turn out to be the same sum of terms $u_i^2v_j^2$ ($i \neq j$) minus cross terms $2u_iu_jv_iv_j$ ($i < j$) — the term-by-term verification is mechanical and is left indicated (exercise 9, with guidance). From there, using $u \cdot v = \|u\|\|v\|\cos\theta$ (Session 3):

$$\|u \times v\|^2 = \|u\|^2\|v\|^2(1 - \cos^2\theta) = \|u\|^2\|v\|^2\sin^2\theta,$$

and since $\theta \in [0, \pi]$, $\sin\theta \geq 0$ and square roots can be taken with no loss. The interpretation: base $\|u\|$ times height $\|v\|\sin\theta$. ■

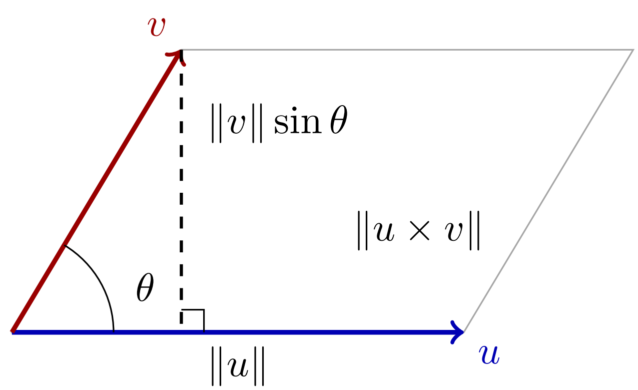

Figure 18: Parallelogram with sides u, v: base $\|u\|$ and height $\|v\|\sin\theta$ (the component of v perpendicular to u); its area is $\|u \times v\|$.

> **The "$u \times v = 0$" case is a limiting case, not a separate property.** Proposition 5.4.3.3 is exactly Theorem 5.4.5 in the limit: $\|u \times v\| = \|u\|\,\|v\|\sin\theta = 0 \iff \sin\theta = 0 \iff u \parallel v$ — the parallelogram flattens and the area drops to zero ($u \times u = 0$ is the extreme case $\theta = 0$). The algebraic proof of 5.4.3.3 with minors remains useful (it uses no trigonometry), but conceptually it *is* the

area hitting zero. **And it also covers what the area argument cannot reach:** if $u = 0$ or $v = 0$ there is no angle θ to speak of, and yet the vectors are proportional and $u \times v = 0$ all the same — that is given by 5.4.3.3, which needs no angles.

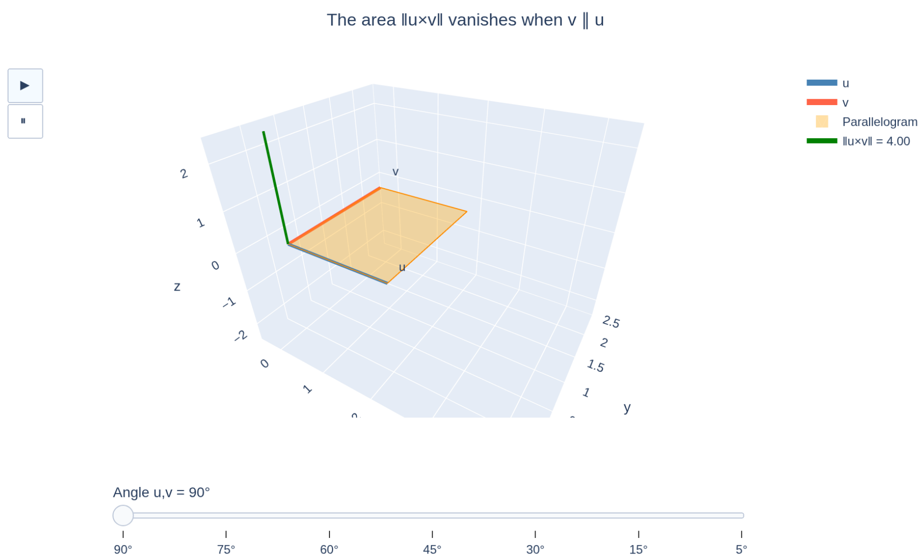

Figure 19: As v rotates: the area $\|u \times v\|$ vanishes when v becomes parallel to u —the parallelogram collapses—, and rotating further to **negative** angles the vector $u \times v$ flips side and points downwards.

[see interactive version]

Orientation: of the two possible perpendicular directions, $u \times v$ points to the side marked by the **turn from** u **towards** v: the screw rule of the figure above. Elsewhere you will almost always meet it as the **right-hand rule** —index u, middle finger v, thumb $u \times v$—, which says exactly the same with a different gesture — the cross product *sees* the orientation, in line with the sign of the determinant (Linear Algebra, Session 6). Antisymmetry ($v \times u = -u \times v$) is that sensitivity made algebra.

Figure 20: The screw rule: turning from u towards v, the screw advances along $u \times v$; turning the other way, it goes down. With $u \parallel v$ it does not move, which is $u \times v = 0$.

[see interactive version]

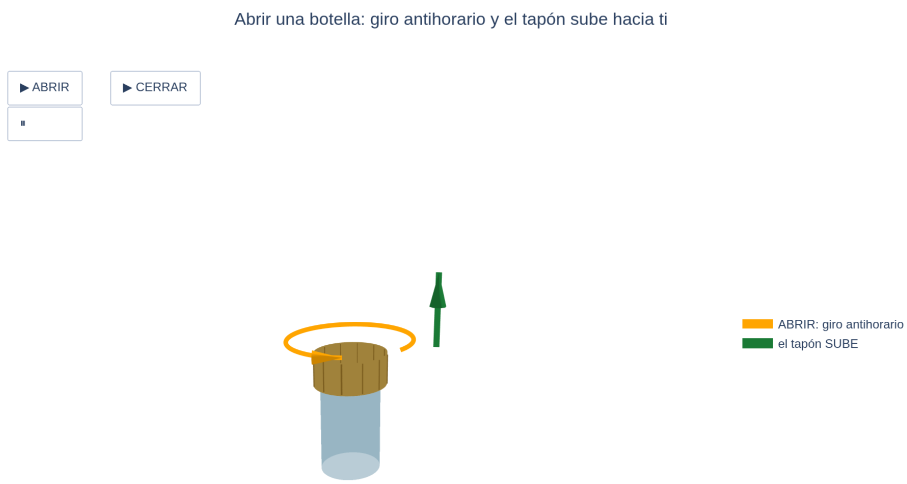

Figure 21: The everyday version: opening a bottle is turning anticlockwise, and the cap comes **up** towards you; closing it is the other way round.

[see interactive version]

> **The anchor, so as not to depend on the gesture.** The rule is pinned down by a single computation in the standard basis: $e_1 \times e_2 = (1,0,0) \times (0,1,0) = (0,0,1) = e_3$ — from x to y comes z. That case decides the sense, and bilinearity (Proposition 5.4.3.2) propagates it to every other pair. Whenever you are unsure which way the thumb points, come back to this example and check it against Definition 5.4.1.

4.2 Scalar triple product and volumes

Definition and Proposition 5.4.6 (scalar triple product = determinant). The **scalar triple product** of $u, v, w \in \mathbb{R}^3$ is $[u,v,w] = u\cdot(v\times w)$, and it coincides with the determinant:

$$[u,v,w] = \det\begin{pmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{pmatrix}.$$

Proof. Expanding the determinant **along the first row** (cofactors):

$$\det = u_1(v_2w_3 - v_3w_2) - u_2(v_1w_3 - v_3w_1) + u_3(v_1w_2 - v_2w_1),$$

which is exactly $u_1(v\times w)_1 + u_2(v\times w)_2 + u_3(v\times w)_3 = u\cdot(v\times w)$ (Definition 5.4.1). *(Here the mnemonic of 5.4.2 shows its truth: the "symbolic determinant" is this expansion with e_i in place of u_i.)* ■

Theorem 5.4.7 (volumes). $|[u, v, w]|$ is the **volume of the parallelepiped** with edges u, v, w. Consequently (Linear Algebra, Theorem 4.6.11):

$$u, v, w \text{ coplanar (dependent)} \iff [u, v, w] = 0.$$

Proof (volume). **Degenerate case first:** if v, w are dependent, then $v \times w = 0$ (Proposition 5.4.3.3), the "base" has area 0, the parallelepiped is flattened (volume 0) and the scalar triple product equals $u \cdot 0 = 0$: the equality holds trivially. **Case v, w independent:** then $n = v \times w \neq 0$ and one can proceed. Base: the parallelogram of v, w, with area $\|n\|$ (Theorem 5.4.5). Height: the component of u perpendicular to the base, which is the projection of u onto the normal n (Proposition 5.4.4): height $= \|\mathrm{proj}_n u\| = \frac{|u \cdot n|}{\|n\|}$ (Session 3 — the division is legitimate: $\|n\| \neq 0$). Volume $=$ base $\times$ height $= \|n\| \cdot \frac{|u \cdot n|}{\|n\|} = |u \cdot (v \times w)|$. *(Fully consistent with "the determinant measures volumes" from Linear Algebra S6 — here we have rebuilt it from metric geometry.)* ■

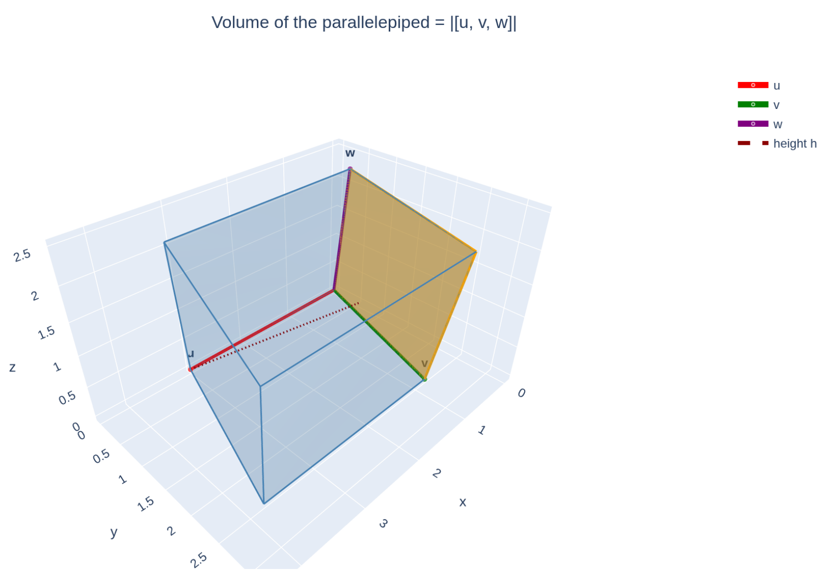

Figure 22: Parallelepiped with edges u, v, w from a common vertex: base the parallelogram of v, w (area $\|v \times w\|$) and height $\|\mathrm{proj}_n u\|$ with $n = v \times w$; if coplanar, it flattens onto a plane (volume 0).

[see interactive version]

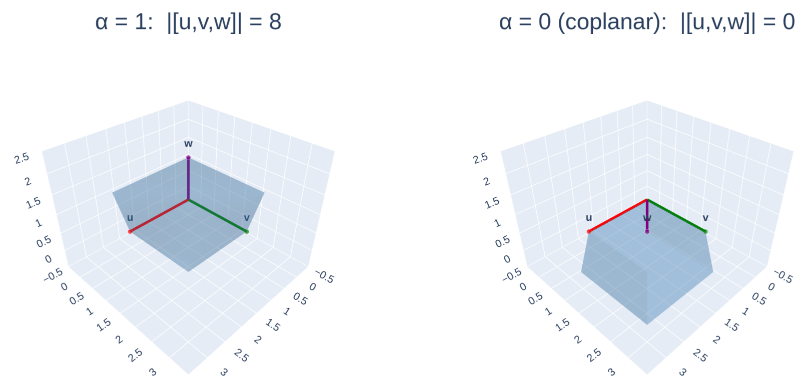

Figure 23: Coplanarity $\iff$ volume 0: as the third vector approaches the plane of the other two, the parallelepiped flattens and $|[u, v, w]| \to 0$.
[see interactive version]

Immediate computational applications: area of a triangle ($\frac{1}{2}\|u \times v\|$), volume of a tetrahedron ($\frac{1}{6}|[u, v, w]|$ — for reference), collinearity and coplanarity tests for points (form the difference vectors and look at the rank/determinant).

The planar case. For $u, v \in \mathbb{R}^2$ there is no cross product, but the area of the parallelogram they determine is

$$\text{area} = \left|\det \begin{pmatrix} u_1 & v_1 \\ u_2 & v_2 \end{pmatrix}\right|,$$

the determinant of the 2×2 matrix carrying them as columns. This is the reading of the determinant as an **area factor** from Linear Algebra (Session 6, §6.1), and it is Theorem 5.4.5 upon embedding $\mathbb{R}^2$ into $\mathbb{R}^3$ as the plane $z = 0$: then $u \times v = (0, 0, u_1v_2 - u_2v_1)$, whose norm is exactly that absolute value.

4.3 Applications

Method 5.4.8 (the normal and the implicit equation of the plane, in one line). Plane through P with direction vectors u, v: its **normal** is $n = u \times v$ (perpendicular to both, 5.4.4, and nonzero by 5.4.3.3 since they are independent). The implicit equation is then

$$n \cdot \vec{PX} = 0 \qquad \text{that is} \qquad n_1x + n_2y + n_3z = n \cdot P,$$

(a point X lies on the plane exactly when $\vec{PX}$ is a direction, that is, perpendicular to the normal — Proposition 5.3.12 in reverse).

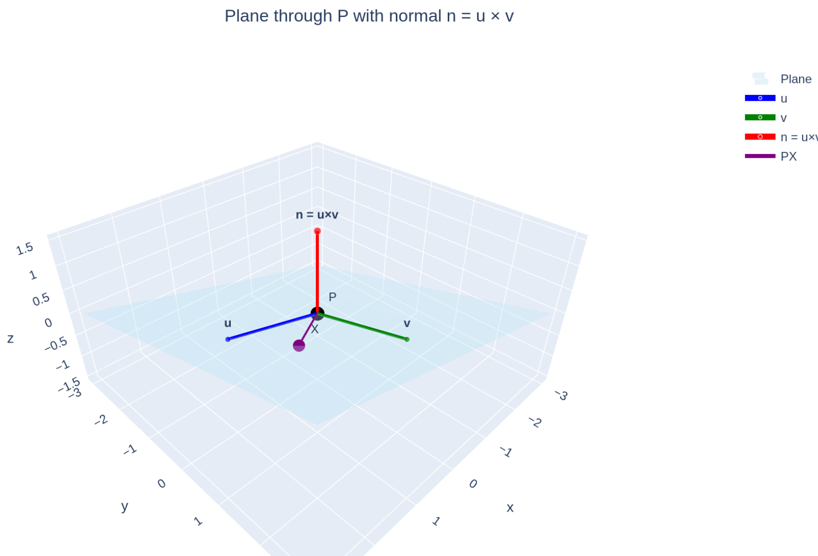

Figure 24: Plane through P with direction vectors u, v and the normal $n = u \times v$ sticking out of the plane; a generic point X with $\vec{PX}$ contained in the plane (perpendicular to n).

[see interactive version]

Theorem 5.4.9 (distance between skew lines). Let $r = P + \operatorname{span}(v)$ and $s = Q + \operatorname{span}(w)$ be skew. Then

$$d(r, s) = \frac{\left|\,[\vec{PQ},\, v,\, w]\,\right|}{\|v \times w\|}.$$

Proof. Since r, s are skew, v, w are independent and $n = v \times w \neq 0$ is **perpendicular to both directions**. For any points $X = P + \alpha v \in r$ and $Y = Q + \beta w \in s$:

$$\vec{XY} = \vec{PQ} - \alpha v + \beta w \quad \Longrightarrow \quad \vec{XY} \cdot n = \vec{PQ} \cdot n$$

(the terms in v and w die against n). That is: **the component along the common normal is the same for every pair of points**.

The bound. By **Cauchy–Schwarz** (Theorem 5.3.5) applied to $\vec{XY}$ and n:

$$|\vec{PQ} \cdot n| = |\vec{XY} \cdot n| \leq \|\vec{XY}\|\,\|n\| \quad \Longrightarrow \quad \|\vec{XY}\| \geq \frac{|\vec{PQ} \cdot n|}{\|n\|}$$

for **every** pair $X \in r$, $Y \in s$: the bound belongs to the two lines, not to two particular points.

And it is attained — which is why it is the distance and not something smaller. Cauchy–Schwarz gives equality **if and only if** $\vec{XY}$ and n are proportional, so it suffices to exhibit one pair with $\vec{XY} \parallel n$. Since v, w are independent and $n = v \times w$ is perpendicular to both, $\{v, w, n\}$ is a basis of $\mathbb{R}^3$: write $\vec{PQ} = av + bw + cn$. Taking $\alpha = a$ and $\beta = -b$ gives

$$\vec{XY} = \vec{PQ} - \alpha v + \beta w = c\,n,$$

proportional to n. That segment, being a multiple of n, is perpendicular to both v and w: its endpoints are the **feet of the common perpendicular**, and those α, β are what locate them. So the minimum exists and equals the bound. Finally $\vec{PQ} \cdot n = \vec{PQ} \cdot (v \times w) = [\vec{PQ}, v, w]$ (Definition 5.4.6). ■

> Note the piece-by-piece assembly: skew lines (S2, determinant criterion), normal (cross product), projection (S3), scalar triple product (5.4.6). **The formula is the whole module condensed.**

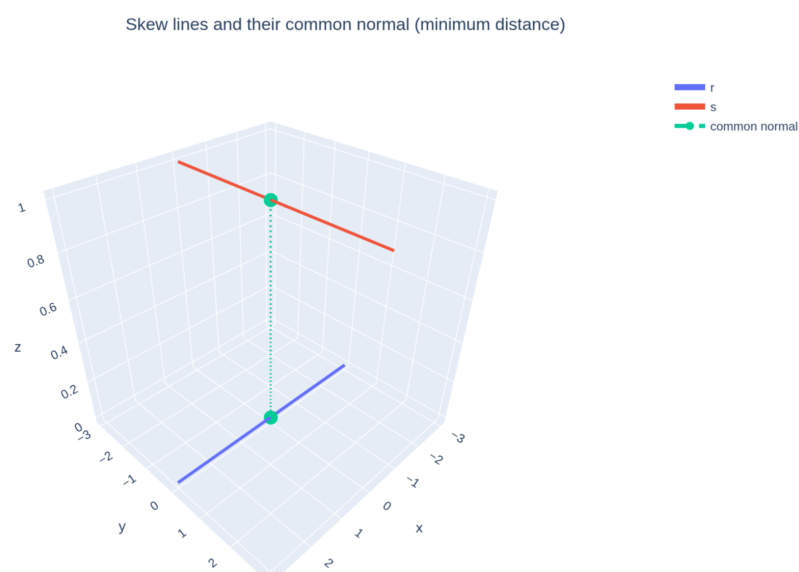

Figure 25: Two skew lines r, s and the segment of the common normal realizing the minimum distance; rotating the view shows that they do not meet.

[see interactive version]

> **Software (RA7) — the normal, in one command.** Machines do these computations too, and it is worth watching them do it. In **GeoGebra 3D**: type `u=(1,0,1)` and `v=(0,1,1)` — the direction vectors of exercise 1 —, ask for `n=Cross(u,v)` and then draw the plane through $P = (1, 1, 0)$ with that normal (exercise 2); rotate the view and check that `n` escapes the plane while `u` and `v` stay inside it. In **Python (sympy)** the same computation fits in three lines:

```python
from sympy import Matrix

u, v = Matrix([1, 0, 1]), Matrix([0, 1, 1])
n = u.cross(v)
print("normal:", n.T)                              # (-1, -1, 1)  ← exercise 1
print("perpendicular?", n.dot(u), n.dot(v))       # 0 0          ← Proposition 5.4.4
print("area of the parallelogram:", n.norm())     # sqrt(3)      ← Theorem 5.4.5
```

Asynchronous activity: reproduce both environments — `Cross` in GeoGebra and `.cross()` in sympy — and the block below, with the data of exercises 1-2 and 3. The arithmetic is the machine's; yours is knowing what each number measures and why it vanishes. The **full statement** —with the predictions to write down before computing, and the four-coplanar-points trap— is in Activity RA7-2 · The normal, in one command. Submission goes through the virtual classroom.

Software (RA7) — the volume, and how it flattens. The determinant of the three vectors gives the volume of the parallelepiped they span. Change the height `h` and run it again: with `h = 0` the three vectors are coplanar and the volume is zero — linear dependence, seen as geometry —, and with `h = -3` the absolute value returns the same volume with the orientation reversed.

```python
from sympy import Matrix

h = 3    # ← the height: change it and re-run. What happens with h = 0? And with h = -3?
u, v, w = [1, 0, 0], [1, 2, 0], [1, 1, h]
print("volume:", abs(Matrix([u, v, w]).det()))
# with h = 0 the three vectors are coplanar: the parallelepiped flattens
```

Exercises

1. ★ With $u = (1, 0, 1)$ and $v = (0, 1, 1)$: compute $u \times v$, check the perpendicularity with the dot product, and find the **area** of the parallelogram and of the triangle they determine.

2. ★ Find the implicit equation of the plane through $P = (1, 1, 0)$ with direction vectors $u = (1, 0, 1)$, $v = (0, 1, 1)$, via the normal (Method 5.4.8). Check with a second point of the plane.

3. ★ Compute the volume of the parallelepiped with edges $(1, 0, 0), (1, 1, 0), (1, 1, 1)$. Then decide: (a) whether the **vectors** $(1, 2, 3), (2, 3, 4), (3, 4, 5)$ — with a common origin — are coplanar (scalar triple product test); (b) whether the **four points** $(0, 0, 0), (1, 2, 3), (2, 3, 4), (3, 4, 5)$ are coplanar *(form the difference vectors from the first one — and note that three points are always coplanar: that is why the test for points needs four).*

4. ★ Compute the **distance between the skew lines** of Example 5.2.6 (r:

the x-axis; $s : (0,0,1) + t(0,1,0)$) with Theorem 5.4.9, and check that the result matches the intuition of the picture ("corridors on different floors").

5. Verify the antisymmetry $v \times u = -(u \times v)$ with Definition 5.4.1 and connect it with the determinant property "swapping rows changes the sign" (via 4.6).

6. Show that $u \times v$ **is not associative**: find u, v, w with $(u \times v) \times w \neq u \times (v \times w)$. *(Hint: try vectors from the standard basis.)*

7. Check that the formal expansion of the "symbolic determinant" (Remark 5.4.2) along the first row reproduces Definition 5.4.1.

8. Follow the minors argument of the proof of 5.4.3.3 with $u = (1,2,4)$ and $v = (2,4,9)$: check that they are independent, compute the **three** 2×2 minors and observe that the first one (columns 1,2) **vanishes** and yet $u \times v \neq 0$ — the argument guarantees *some* nonzero minor, not the first one.

9. **(Lagrange, guided.)** Expand $\|u \times v\|^2$ and $\|u\|^2\|v\|^2 - (u \cdot v)^2$ and check that both equal $\sum_{i<j}(u_i v_j - u_j v_i)^2$.

10. **(Module synthesis.)** For $r : (1,0,0) + t(1,1,0)$ and $s : (0,1,1) + t(1,-1,1)$: decide their position (S2), and if they are skew compute their distance (5.4.9); if not, whichever distance applies (S3).

11. ★★ **(The cross product formula, derived — not handed over.)** Look for the $w = (w_1, w_2, w_3)$ perpendicular to both u and v, with u, v independent.

    (a) Write the two conditions $\langle w, u\rangle = 0$, $\langle w, v\rangle = 0$ as a **homogeneous linear system** of 2 equations in 3 unknowns, with matrix $\begin{pmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{pmatrix}$.
    (b) Check that its rank is 2 (this is where independence of u, v is used) and deduce, by Rouché–Frobenius, that its solutions form a subspace of dimension $3 - 2 = 1$: **a line** of candidates.
    (c) Check that $u \times v$ solves the system, and conclude that **every** solution is a multiple of it.
    (d) What is left to fix is size and direction: add the equation $\|w\| = \|u\|\,\|v\| \sin\theta$ (the area of the parallelogram) and observe that it leaves **exactly two** candidates, opposite to each other; the **orientation** convention ($e_1 \times e_2 = e_3$) picks one. *(Moral: the formula in Definition 5.4.1 is not a whim. It is the only way out of demanding perpendicularity, area and orientation.)*

Going further and references

Going further (why only in $\mathbb{R}^3$?). A bilinear antisymmetric product that returns a *vector in the same space*, perpendicular to the factors and with the norm of the area, exists only in dimension 3 (and, surprisingly, in dimension 7, via the octonions). In general dimension, the natural object is not a vector but the determinant/alternating forms — the cross product is a happy accident of $\mathbb{R}^3$. Introductory reference: Stillwell, *Naive Lie Theory*, ch. 1, or Massey's classic survey, *Cross products of vectors in higher dimensional Euclidean spaces.*

Going further (isometries — the module's additional material). The **isometries** are the transformations that preserve distances: translations, rotations, reflections and their compositions. The linear ones have matrices with orthonormal columns, and their determinant is ± 1: $+1$ preserves orientation (rotations), -1 reverses it (reflections) — the closing clasp with Session 6 of Linear Algebra. See Audin, *Geometry*, ch. II–III, or de Burgos.

References. Cross product and scalar triple product: Hernández, *Álgebra y geometría*; de Burgos. Volumes and determinants: Strang, §5.3; the Linear Algebra notes, Session 6. Distance between skew lines: de Burgos.